

ECO101 - Introduction to Microeconomics

Department of Economics and Social Science (ESS)
BRAC University

Elasticity

Lecture 3

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 4
- Principles of Economics by Mankiw, 6th Edition
 - Chapter: 5

Outline

- Define, calculate, and explain the factors that influence the price elasticity of demand
- Define, calculate, and explain the factors that influence the cross elasticity of demand and the income elasticity of demand
- Define, calculate, and explain the factors that influence the elasticity of supply

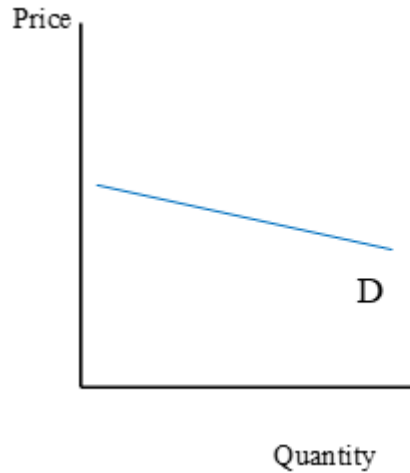
What is Elasticity?

- In last lecture, we noted that consumers usually buy more of a good when its price is lower, when their incomes are higher, when the prices of substitutes for the good are higher, or when the prices of complements of the good are lower
- Elasticity is a measure of the responsiveness of quantity demanded or quantity supplied to a change in one of its determinants

Elasticity of Demand

Elasticity

How sensitive is quantity demanded to a change in price?



Why is it useful understanding elasticity?

- Businesses might be interested in the effect of a price change on quantity sold and how it would impact the profits
- Governments might want to know how behaviour changes when a specific policy is put in place
 - Example: carbon tax, road tolls, change in minimum wage

Price Elasticity of Demand

- Price elasticity of demand is a measure of the responsiveness of the quantity demanded of a good to a change in its price when all other influences on buying plans remain the same
- We calculate price elasticity of demand as the percentage change in quantity demanded divided by percentage change in price

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}.$$

Price Elasticity of Demand - Example 1

Suppose that a 10 percent increase in the price of an ice cream cone causes the amount of ice cream you buy to fall by 20 percent

Calculate your price elasticity of demand

Price elasticity of demand : $20 \text{ percent} / 10 \text{ percent} = 2$

This implies that the change in the quantity demanded is proportionately twice as large as the change in the price

Price Elasticity of Demand - Example 2

Suppose initially the price of pizza is \$20.50 and at this price 9 pizzas an hour are demanded. Then the price falls to \$19.50 a pizza and the quantity demanded increases to 11 pizzas an hour

$$PED = \frac{\frac{11 - 9}{9} \times 100}{\frac{19.50 - 20.50}{20.5} \times 100} = 4.5$$

If the price rises from \$19.5 to \$20.5, price elasticity of Demand = 3.6

Price Elasticity of Demand - Example 2 (Midpoint Method)

Using Midpoint Method

Change in price is expressed as a percentage of the average price and the change in the quantity demanded is expressed as a percentage of the average quantity

$$\text{Price elasticity of demand} = \frac{\frac{\Delta Q}{Q_{ave}} \times 100}{\frac{\Delta P}{P_{ave}} \times 100} = \frac{Q_F - Q_0}{P_F - P_0} \times \frac{P_{ave}}{Q_{ave}}$$
$$\Delta P = P_F - P_0$$
$$\Delta Q = Q_F - Q_0$$
$$P_{ave} = \frac{P_F + P_0}{2}$$
$$Q_{ave} = \frac{Q_F + Q_0}{2}$$

Price Elasticity of Demand - Example 2 (Midpoint Method)

$$\Delta P = P_F - P_0 = \$20.5 - \$19.5 = \$1$$

$$\Delta Q = Q_F - Q_0 = 9 - 11 = -2$$

$$P_{ave} = \frac{P_F + P_0}{2} = (20.5 + 19.5)/2 = 20$$

$$Q_{ave} = \frac{Q_F + Q_0}{2} = (11 + 9)/2 = 10$$

$$\text{Price elasticity} = \frac{\% \Delta Q}{\% \Delta P} = \frac{\frac{\Delta Q}{Q_{ave}} \times 100}{\frac{\Delta P}{P_{ave}} \times 100} = \frac{Q_{ave}}{P_{ave}} \times \frac{Q_F - Q_0}{P_F - P_0} = 20/5 = 4$$

Price Elasticity of Demand

- So to compare price elasticities of demand, we use the absolute value and ignore the minus sign
- Elasticity is independent of the units of measurement
- Elasticity is based on proportional changes

Price Elasticity of Demand - Example 3

Suppose we have demand curve as such

$$Q = 12 - 2P$$

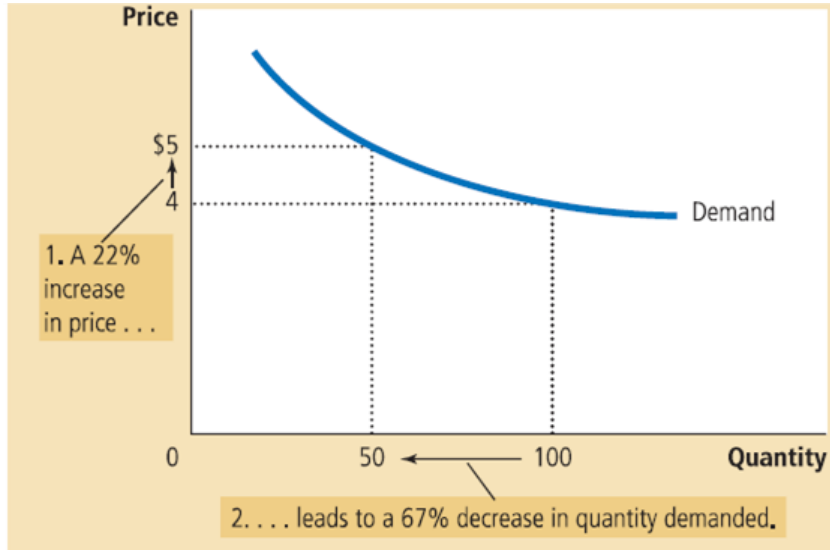
Find the price elasticity of demand when the price changes from \$1 to \$2.

Answer:

Price elasticity of demand = 0.33

Inelastic and Elastic Demand

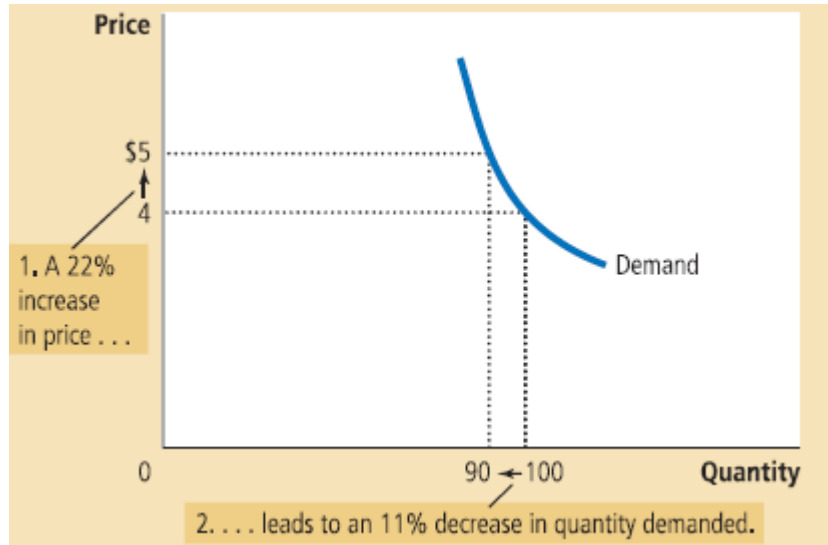
Demand is considered to be **elastic** when the elasticity is greater than 1



- Percentage change in quantity demanded exceeds the percentage change in price
- Example: automobiles and furniture

Inelastic and Elastic Demand

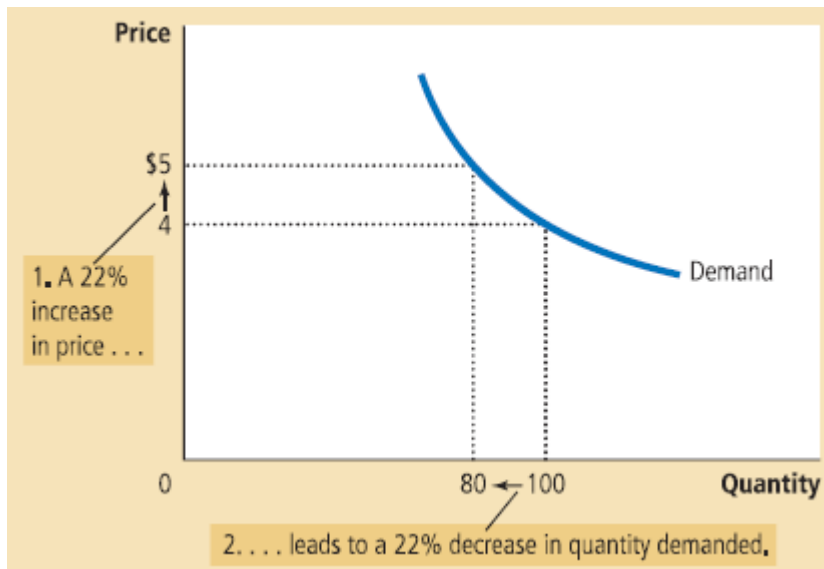
Demand is considered to be **inelastic** when the elasticity is less than 1



- In other words, percentage change in the quantity demanded is less than the percentage change in price
- In this case the price elasticity of demand is between 0 and 1
- Example: food shelter

Inelastic and Elastic Demand

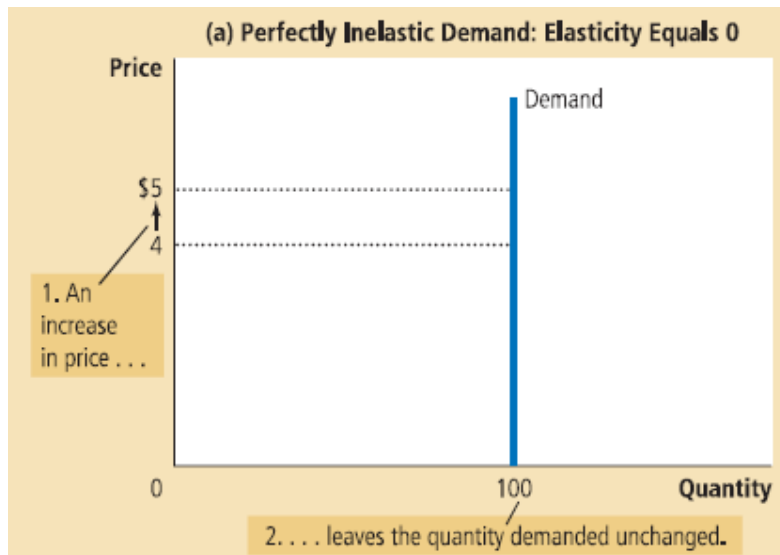
If the elasticity is exactly 1, the demand is said to have **unit elasticity**



- Percentage change in quantity demanded equals percentage change in the price

Inelastic and Elastic Demand

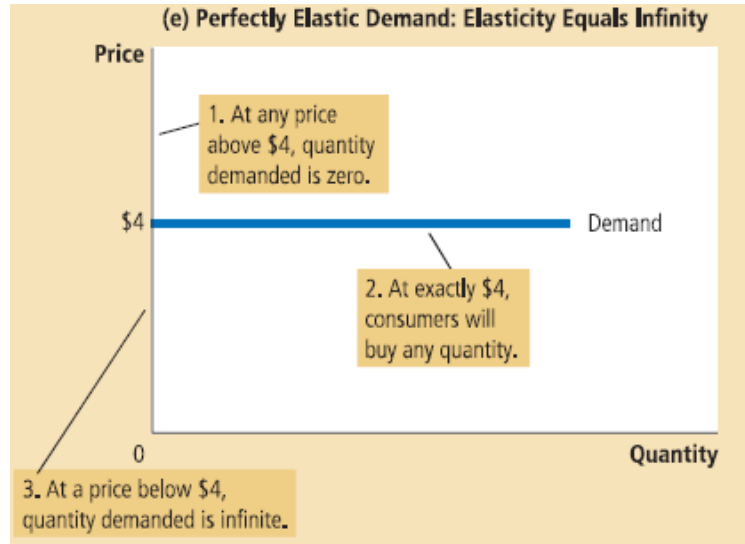
Demand is said to be **perfectly inelastic** when the price elasticity of demand is zero



- The quantity demanded remains constant when price changes
- Example: insulin (diabetic patients do not tend to change the quantity of insulin they buy in a month due to a fall or rise in price)

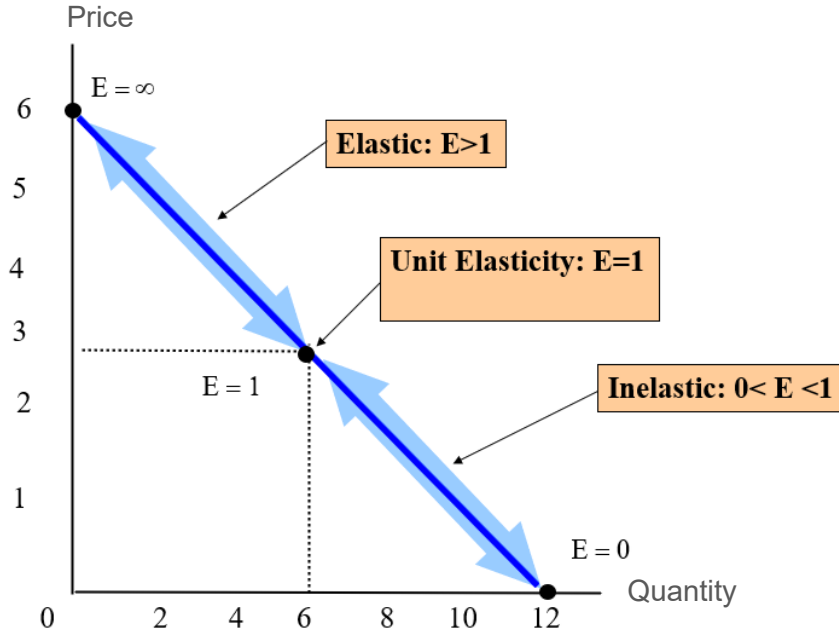
Inelastic and Elastic Demand

If the quantity demanded changes by an infinitely large percentage in response to a tiny price change, then the price elasticity of demand is infinity and the good is said to have **perfectly elastic demand**



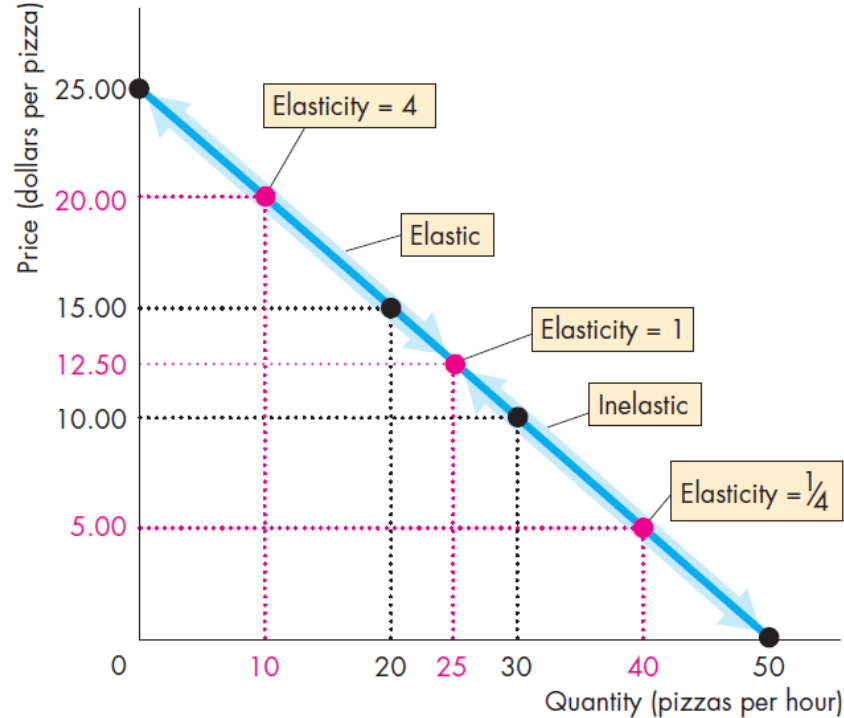
- Example: soft drink from two machines located side by side (perfect substitutes)

Elasticity along the Demand Curve



➤ A linear demand curve has a constant slope but a varying elasticity

Elasticity along the Demand Curve - Example

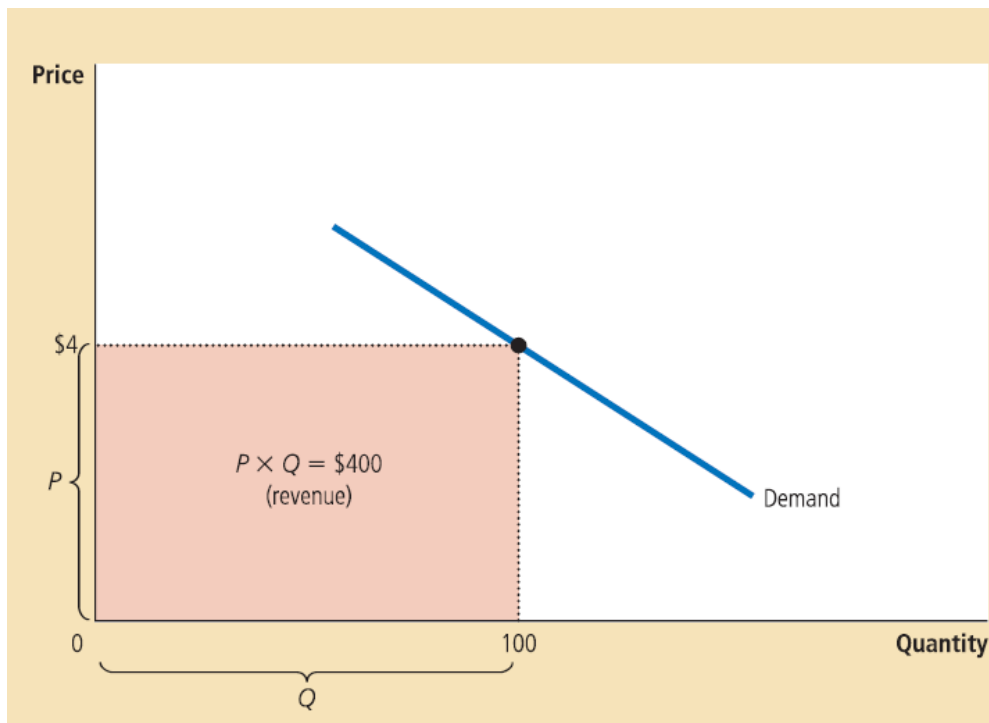


Total Revenue and Elasticity

- When studying changes in supply or demand in a market, one variable we often want to study is total revenue, the amount paid by buyers and received by sellers of the good
- Total revenue = $P \times Q$ (*the price of the good times the quantity of the good sold*)
- Total revenue can be shown graphically

Total Revenue and Elasticity

Figure 2



Total Revenue

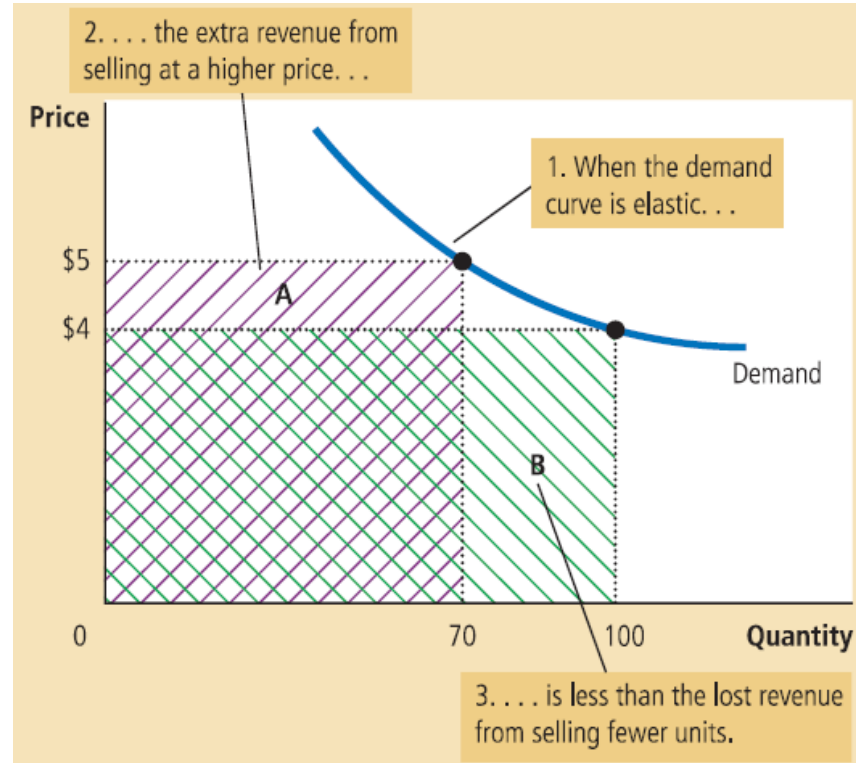
The total amount paid by buyers, and received as revenue by sellers, equals the area of the box under the demand curve, $P \times Q$. Here, at a price of \$4, the quantity demanded is 100, and total revenue is \$400.

Total Revenue and Elasticity

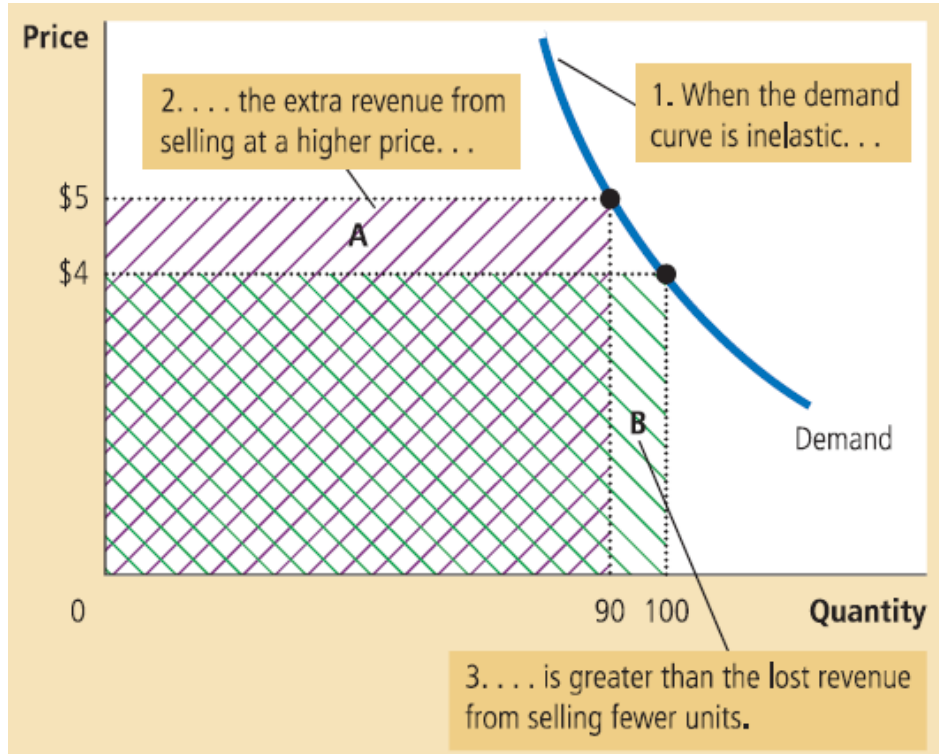
The change in total revenue depends on the elasticity of demand in the following way

- If the demand is elastic, a 1 percent price cut increases the quantity sold by more than 1 percent and total revenue increases
- If demand is inelastic, a 1 percent price cut increases the quantity sold by less than 1 percent and total revenue decreases
- If the demand is unit elastic, a 1 percent price cut increases the quantity sold by 1 percent and total revenue is unchanged

Total Revenue and Elasticity - Case of Elastic Demand



Total Revenue and Elasticity - Case of Inelastic Demand



Total Revenue Test

We can use the relationship between elasticity and total revenue to estimate elasticity using the total revenue using total revenue test

- If a price cut increases total revenue, demand is elastic
- If a price cut decreases total revenue, demand is inelastic
- If a price cut leaves total revenue unchanged, demand is unit elastic

Factors that influence the Elasticity of Demand

The elasticity of demand for a good depends on the following

- **Closeness of substitutes:** The closer the substitutes for a good or service, the more elastic is the demand for it

Necessities vs Luxuries:

- A necessity has poor substitutes and is crucial for our wellbeing. So a necessity generally has an inelastic demand. Example: prescribed medications
- A luxury usually has many substitutes, one of which is not buying it and generally has an elastic demand. Example: yacht

Factors that influence the Elasticity of Demand

- **Proportion of income spent on the Good:** Other things remaining the same, the greater the proportion of income spent on a good, the more elastic (or less inelastic) is the demand for it
 - Housing takes a large proportion of your budget, while gums take a tiny proportion
- **Time elapsed since price changed:** The longer the time that has elapsed since a price change, the more elastic is demand
 - For example, how the demand for oil became more elastic as more time elapsed following huge price hike

Cross Elasticity of Demand

- The **cross elasticity of demand** is a measure of the responsiveness of the demand for a good to a change in the price of a substitute or complement, other things remaining the same

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good 1}}{\text{Percentage change in the price of good 2}}.$$

- The cross elasticity of demand can be positive or negative - it is positive for a substitute and negative for a complement

Cross Elasticity of Demand - Example 1

Complements

Suppose that the price of pizza is constant and people buy 11 pizzas an hour. Then the price of soft drinks rises from \$1.50 to \$2.50. No other influence on buying plans changes and the quantity of pizzas bought falls to 9 an hour.

Calculate cross elasticity of demand for pizza.

$$XED = -0.4$$

- Because pizzas and soft drinks are complements, when the price of a soft drink rises, the demand for pizza decreases

Cross Elasticity of Demand - Example 2

Substitutes

Suppose that the price of pizza is constant and people buy 9 pizzas an hour. Then the price of burger rises from \$1.50 to \$2.50. No other influence on buying plans changes and the quantity of pizzas bought increases to 11 an hour.

Calculate cross elasticity of demand for pizza.

- $XED = 0.4$
- Because pizzas and burgers are substitutes, when the price of a burger rises, the demand for pizza increases

Cross Elasticity of Demand

- If two items are close substitutes, such as two brands of spring water, the cross elasticity is large
- If two items are somewhat unrelated to each other, such as newspapers and orange juice, the cross elasticity is small - perhaps even zero

Cross Elasticity of Demand

Suppose: $Q_d = 800 - 20P_s + 10P_p$

Where: Q_d = Quantity demanded of sparkling water (L);

P_s = Price of sparkling water (\$); P_p = Price of plain bottled water(\$)

Assume initially: $P_s = \$3$ & $P_p = \$2$

*Now suppose the price of plain bottled water increases from \$2 to \$2.50
(while the price of sparkling water stays at \$3).*

*Calculate and interpret the **cross – price elasticity of demand (XED)**.*

Income Elasticity of Demand

- **Income elasticity of demand** is a measure of the responsiveness of the demand for a good or service to a change in income, other things remaining the same

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}.$$

- It measures how the quantity demanded changes as consumer income changes

Income Elasticity of Demand

- Income elasticities of demand can be positive or negative and they fall into these interesting ranges:
 - Greater than 1 (normal good, income elastic)
 - Positive and less than 1 (normal good, income inelastic)
 - Negative (inferior good)

Income Elastic Demand

- Suppose that the price of pizza is constant and 9 pizzas an hour are bought
- Then income rise from \$975 to \$1025 a week
- No other influence on buying plans changes and the quantity of pizzas sold increases to 11 an hour

Calculate the income elasticity of demand for pizza.

Income elasticity of demand for pizza is 4

Income Elasticity of Demand

Suppose: $Q_d = 1200 - 30P + 0.8Y$

Where: Q_d = Quantity demanded of organic coffee (cups);

P = Price; Y = Consumer Income

Assume initially: $P = \$4$ and $Y = \$3000$

Now suppose consumer income increases from \$3,000 to \$3,400.

*Calculate and interpret the **income elasticity of demand (YED)***

Elasticity of Supply

Elasticity of Supply

- The elasticity of supply measures the responsiveness of the quantity supplied to a change in the price of a good when all other influences on selling plans remain the same
- Price elasticity of supply is calculated using the following formula:

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}.$$

Elasticity of Supply - Example 1

Suppose that an increase in the price of milk from \$2.85 to \$3.15 a gallon raises the amount that dairy farmers produce from 9,000 to 11,000 gallons per month.

Calculate the elasticity of supply.

Answer:

Price elasticity of supply = 2

In this example, the elasticity of 2 indicates that the quantity supplied changes proportionately twice as much as the price

Elasticity of Supply - Example 2

Consider we have the following supply curve:

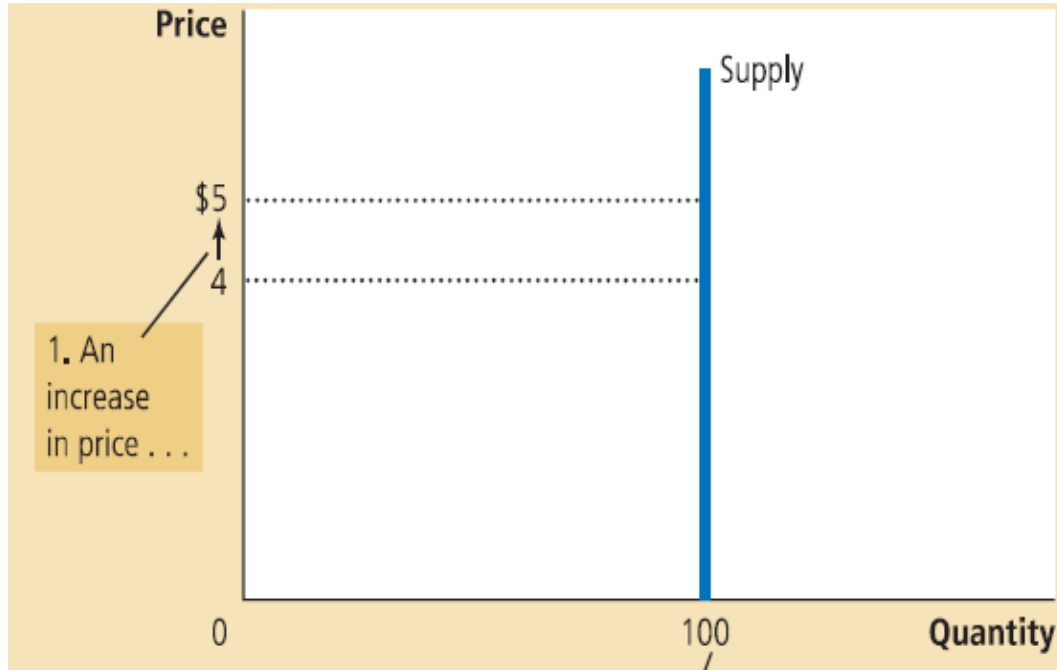
$$Q = P - 20$$

Calculate the elasticity of supply when price is equal \$40 and \$60

Answer:

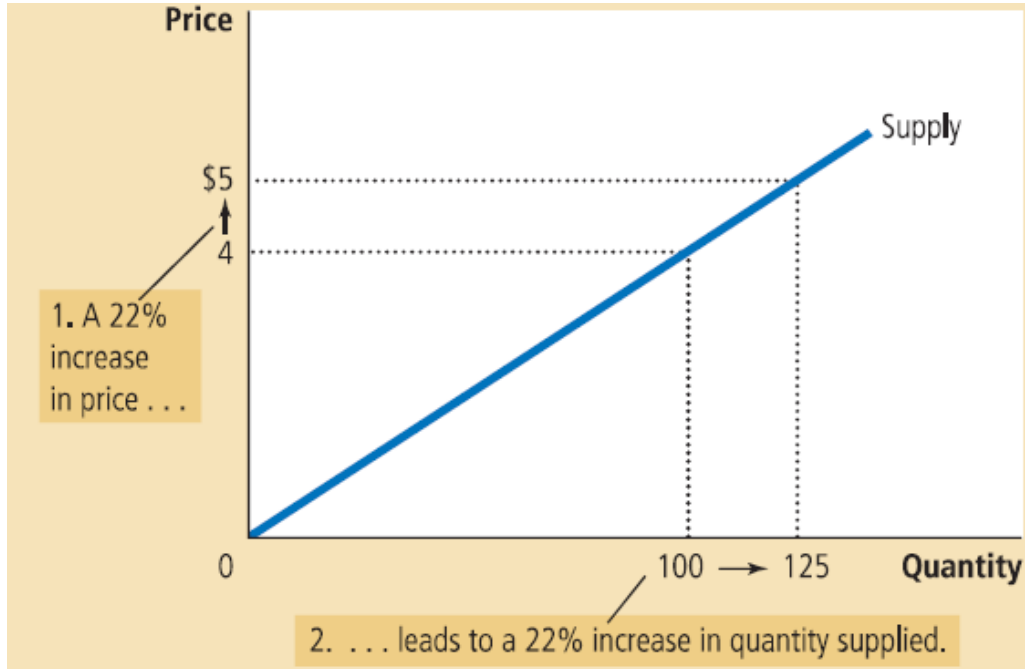
Price elasticity of supply = 1.7

Inelastic and Elastic Supply



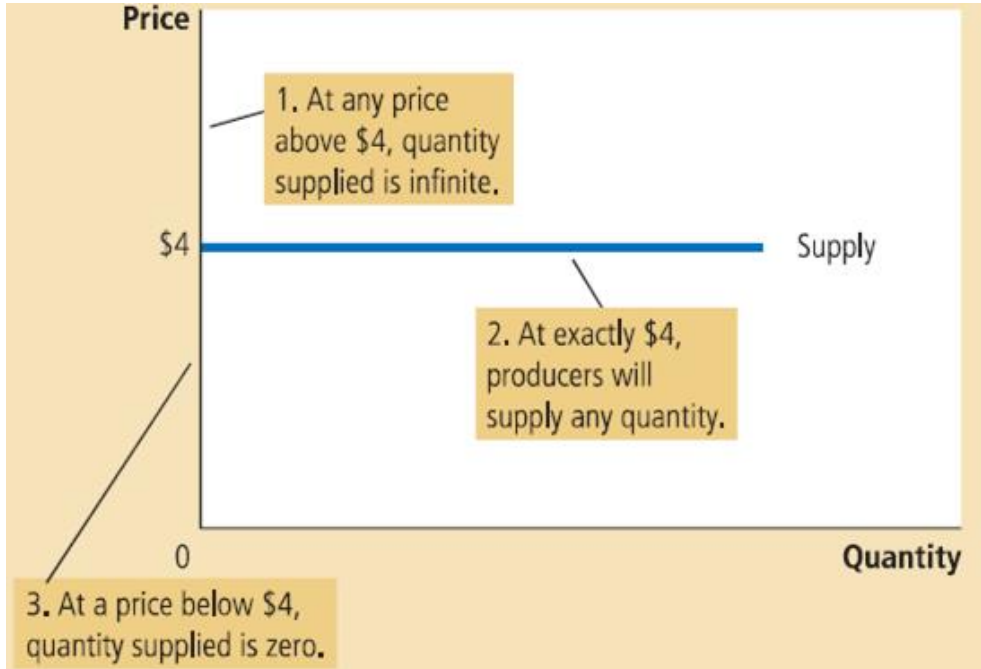
- In this case, elasticity is equal to zero and the supply curve is **perfectly inelastic**
- The quantity supplied is fixed regardless of the price

Inelastic and Elastic Supply



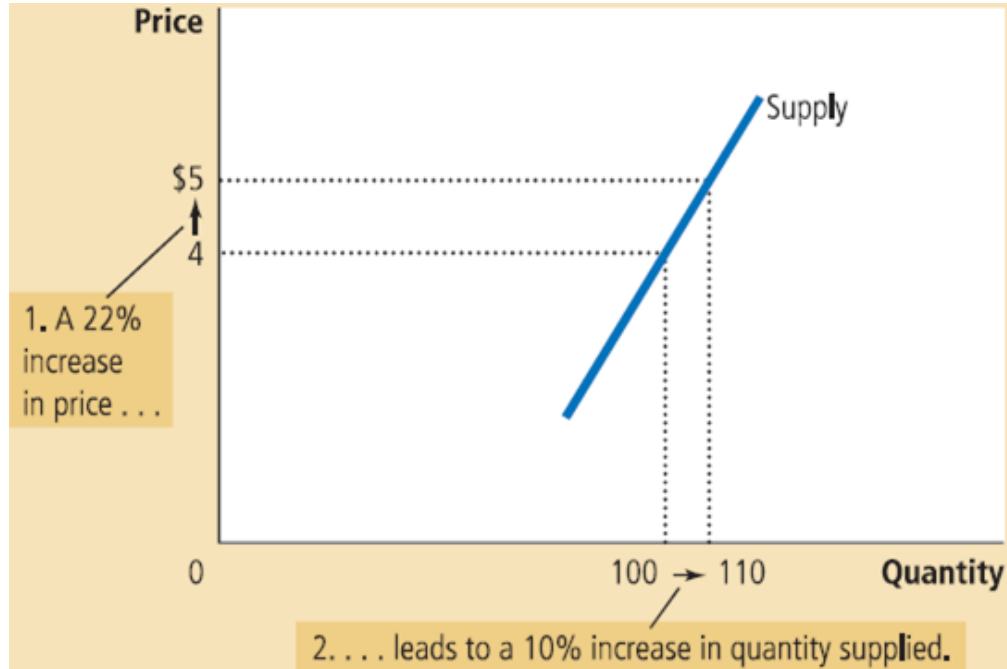
- When elasticity equals 1, the supply is said to be **unit elastic**
- Percentage change in price equals percentage change in quantity

Inelastic and Elastic Supply



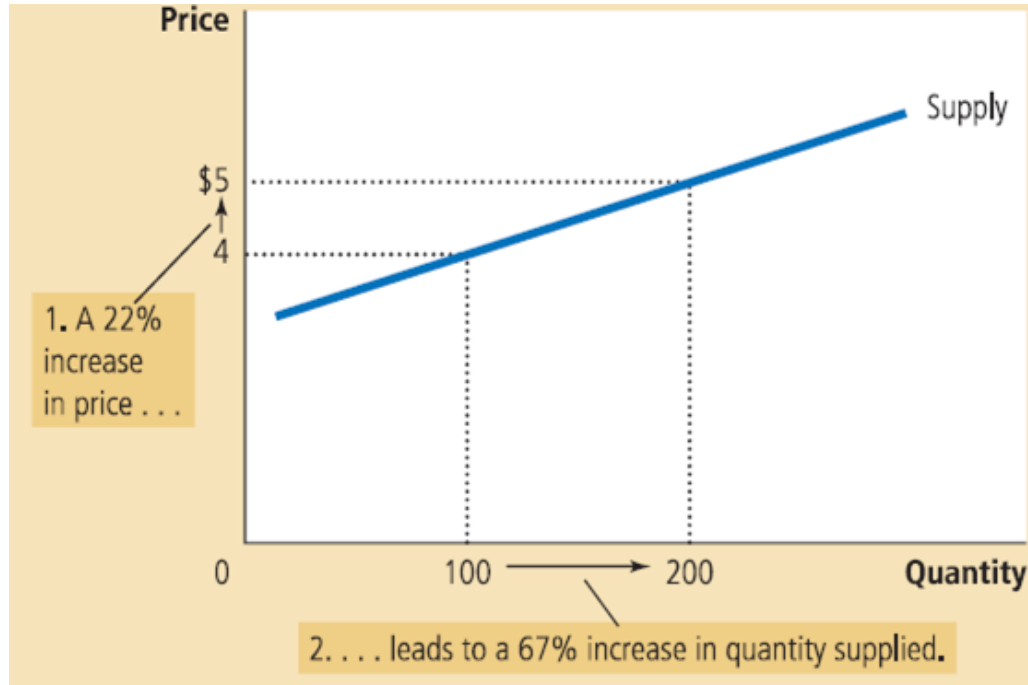
- A supply curve is said to be **perfectly elastic** as the price elasticity of supply approaches infinity and the supply curve becomes horizontal
- Very small changes in the price lead to very large changes in the quantity supplied

Inelastic and Elastic Supply



- A supply is said to be **inelastic** when the price elasticity of supply is less than 1

Inelastic and Elastic Supply



- A supply is said to be **elastic** when the price elasticity of supply is greater than 1

Factors that Influence the Elasticity of Supply

➤ **Resource substitution possibilities:**

- Some goods and services can be produced only by using unique or rare productive resources. These items have a low or even zero elasticity of supply
- Example: a Picasso painting, sand for making silicon chip
- Other goods and services can be produced by using commonly available resources that could be allocated to a wide variety of alternative task. Such items have high elasticity of supply
- Example: wheat can be grown on land that is almost equally good for growing corn

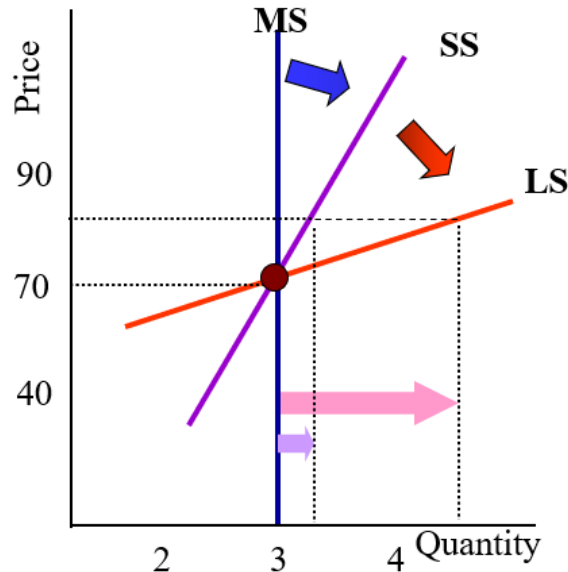
Factors that Influence the Elasticity of Supply

- **Time frame for the supply decision:** To study the influence of the amount of time elapsed since a price change we distinguish three time frames of supply
 - Momentary supply
 - Short-run supply
 - Long-run supply

Factors that Influence the Elasticity of Supply

- Time frame for the supply decision:

Price Elasticity of Supply



Mathematical problems discussed and/or demonstrated during the lecture